

I BURNS MANAGEMENT

Acute burns = thermal (flame and smoke), chemical, electrocution, or radiation.
Scene safety is paramount in these patients. Follow CBRN guidelines for chemical and radiation, and isolate power for electrocution.

EXTERNAL MAJOR HAEMORRHAGE CONTROL

AIRWAY ASSESSMENT AND MANAGEMENT

CHECK, CLEAR AND MAINTAIN AIRWAY

U or HYPOXIA	ADJUNCTS or ADVANCED AIRWAY MANAGEMENT
RESPONSE TO PAIN	CONSIDER ADJUNCTS or BASIC MANOEUVRES
RESPONSE TO VOICE	CONSIDER NPA or BASIC MANOEUVRES/POSITION

AIRWAY or FULL THICKNESS FACIAL BURNS and:

STRIDOR, HOARSE VOICE, INTRA-ORAL OEDEMA, SINGED NASAL HAIR, CARBONACEOUS SPUTUM or RESPIRATORY DISTRESS/FAILURE

CONSIDER EARLY PHEA FOR DEFINITIVE AIRWAY or
PERFORM SURGICAL AIRWAY IN SEVERE SWELLING

BREATHING ASSESSMENT AND MANAGEMENT

APPLY OXYGEN	REDUCED CONSCIOUSNESS, SIGNS OF CARBON MONOXIDE POISONING or HYPOXIC (SPO ₂ <94%). CONSIDER IN PATIENTS THAT HAVE BEEN BURNING PLASTIC = CYANIDE
CIRCUMFERENTIAL TORSO BURNS	CONSIDER SEDATION/ANALGESIA IF FEASIBLE AND PERFORM ESCHAROTOMY RB CONSIDER EARLY REQUEST FOR MEDEVAC or TRAINED CLINICIAN TO ATTEND SCENE
CHEMICAL	SOME CHEMICALS = PULMONARY IRRITATION/WHEEZE CONSIDER SALBUTAMOL NEB
BLATOMFC	ASSESS AND TREAT LIFE-THREATENING THORACIC INJURIES AS NORMAL

CIRCULATION ASSESSMENT AND MANAGEMENT

ASSESS/MANAGE SHOCK	CHECK RADIAL PULSES, ECG MONITORING (IF ABLE) AND REGULAR BPs. GAIN TWO FORMS OF IV ACCESS (VIA BLE SKIN ONLY) or IO ACCESS ADMINISTER BLOOD IF INITIALLY SHOCKED FROM HAEMORRHAGE/POLYTRAUMA
ELECTROCUTION	CONTINUOUS AND 12 LEAD ECG FOR DYSRHYTHMIA IF CHEST PAIN/PALPITATIONS, LOSS OF CONSCIOUSNESS or RESOLVED CARDIAC ARREST
CIRCUMFERENTIAL ABDO/LIMB BURNS	CONSIDER ESCHAROTOMY RB FOR CIRCUMFERENTIAL ABDO BURNS ASSESS FOR COMPARTMENT SYNDROME or DEAD LIMB - EVACUATE IMMEDIATELY

BURN RESUSCITATION FLUID (BRF)

ASSESS % OF BURNT TBSA	>20% TBSA IN ADULTS REQUIRES FLUID RESUSCITATION (>10% IN CHILDREN)	
PARKLAND FORMULA (BRF)	PATIENT'S WEIGHT (KG) x %TBSA x 4 = FLUID (ml)	
	1 st HALF OVER 8 HOURS	2 nd HALF OVER 16 HOURS
EFFECTIVE BRF	MAINTAIN URINARY OUTPUT 1ml/kg/hr AS MINIMUM URINARY CATHETER IS ADVISED IN PROLONGED CARE.	

DEAD LIMB

COLD, PULSELESS, SEVERE PALLOR, TENSELY SWOLLEN, NO SENSORY OR MOTOR FUNCTION

COMPARTMENT SYNDROME

PAIN (SEVERE), PARAESTHESIA, POIKILOthermia, PALLOR, PARALYSIS, PULSELESSNESS

DISABILITY ASSESSMENT AND ENVIRONMENTAL CONSIDERATIONS

HEAD INJURY	CHECK GCS AND PUPILS FOR POSSIBLE HEAD INJURY IN POLYTRAUMA OR HYPOXIA
ANALGESIA AND DRESSINGS	ADMINISTER ANALGESIA AS SOON AS FEASIBLE AND PLACE CLINGFILM OVER BURNS AVOIDING WRAPPING DUE TO SWELLING. REMOVE ANY JEWELLERY IF POSSIBLE
HEAT AND HANDLE	COOL THE BURN WITH COOL DRESSINGS BUT KEEP THE PATIENT WARM USING HPMK
OTHER MEDICATIONS	CONSIDER ANTIBIOTICS IF THE PATIENT HAS CONTAMINATED WOUNDS OR OTHER INJURIES. TXA IS ONLY INDICATED IF BLEEDING HAS OCCURRED OR HEAD INJURY.

CONVEY TO APPROPRIATE BURNS UNIT OR ROLE 2/3 WITH SURGICAL CAPABILITY IN ISOLATED BURNS.
IF POLYTRAUMATIC LIFE-THREATENING INJURIES CONVEY TO ROLE 2 OR MAJOR TRAUMA CENTRE INSTEAD.

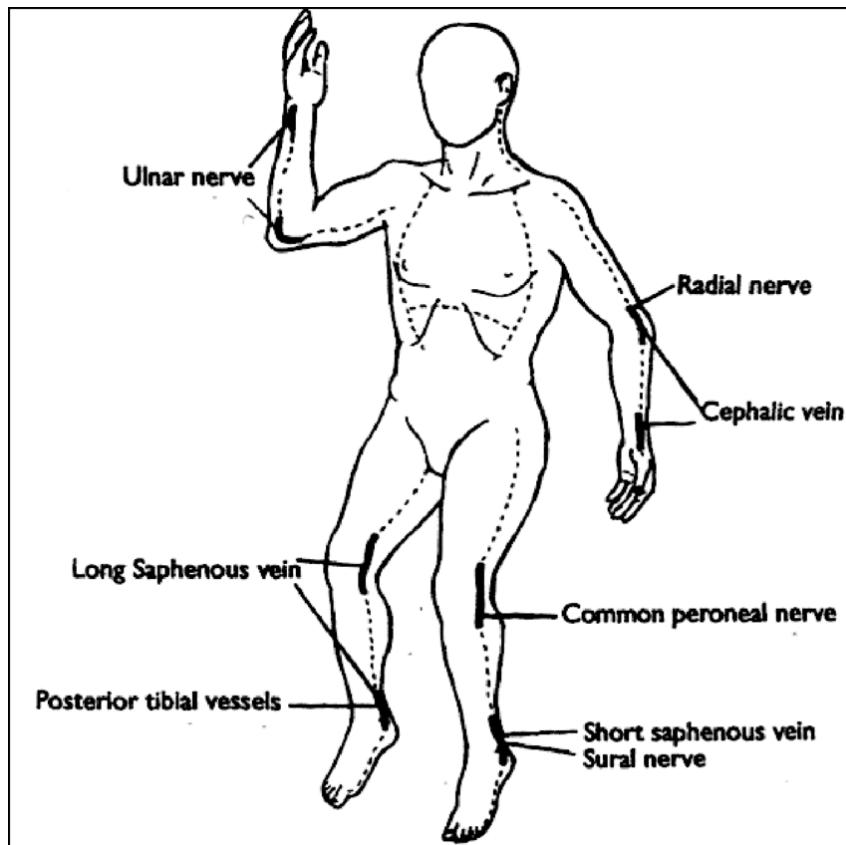


Diagram of Escharotomy Lines for Torso, Neck or Limb Circumferential Burns

POINTS TO NOTE

1. BURN INJURIES ARE LIKELY TO COME WITH CONCOMITANT TRAUMA, SUCH AS FALL FROM HEIGHT AFTER ELECTROCUTION OR BLAST INJURIES. TREAT ACCORDINGLY USING THE PRIMARY SURVEY.
2. HAVE A LOWER THRESHOLD TO PROVIDE DEFINITIVE AIRWAY SUPPORT FOR AIRWAY BURNS WITH ANY OF THE RED FLAGS NOTED ABOVE. DO NOT WAIT UNTIL THE PATIENT HAS SEVERE SWELLING OR INEFFECTIVE BREATHING. AIRWAY COMPROMISE IS AN ABSOLUTE INDICATION FOR PREHOSPITAL EMERGENCY ANAESTHESIA (PHEA).
3. CIRCUMFERENTIAL BURNS TO THE TORSO CAN RESTRICT PATIENTS BREATHING AND CAUSE A TOURNIQUET EFFECT ON THE ABDOMEN, ESCHAROTOMY IS NEEDED AND SHOULD BE DONE BY SUITABLY TRAINED CLINICIANS. DO NOT DELAY CONVEYANCE TO SURGICAL UNIT.
4. OXYGEN SHOULD ONLY BE ADMINISTERED ONCE THE PATIENT IS AWAY FROM FLAME OR SMOKE.
5. BURNS RESUSCITATION FLUID DOES NOT INCLUDE RESUSCITATION FLUID, MAINTENANCE FLUID OR BLOOD REQUIRED FOR ACUTE SHOCK. MANAGE THE PATIENT'S CARDIOVASCULAR INSTABILITY FIRST, ESPECIALLY IN POLYTRAUMA PATIENT WHERE HAEMORRHAGE IS A CONCERN.
6. THERE IS CONCERN FOR COMPARTMENT SYNDROME IN CIRCUMFERENTIAL LIMB BURNS. ESCHAROTOMY OF LIMBS SHOULD BE AVOIDED IN THE PREHOSPITAL SETTING IF POSSIBLE.
7. THESE PATIENTS ARE LIKELY TO REQUIRE MULTIMODAL ANALGESIA APPROACH FOR SEVERE PAIN.

Guideline written in accordance with SFM Clinical Standard Operating Guideline 011 'Burns' (2022)

References

- Battaloglu, E., et al. 2020. Faculty of Pre-Hospital Care & British Burn Association Expert Consensus Meeting: Management of Burns in Pre-Hospital Trauma. Available at: [2020-09-20-burns-consensus.pdf \(rcsed.ac.uk\)](https://www.rcsed.ac.uk/2020-09-20-burns-consensus.pdf)
- American Burn Association. 2018. Advanced Burn Life Support Course Manual. Available at: [2018-abls-providermanual.pdf \(ameriburn.org\)](https://www.ameriburn.org/2018-abls-providermanual.pdf)
- NICE. 2023. *Burns and Scalds*. Available at: [Burns and scalds | Health topics A to Z | CKS | NICE](https://www.nice.org.uk/guidance/NG194)